PLOS ONE submission fields

Prepared for the 18 October 2026 manuscript update. The author-reported existing submission is **PONE-D-26-50597, submitted 9 October 2026**, as already recorded in the historical candidate materials. These fields accompany an author-prepared update candidate. No authenticated portal or editorial invitation has been inspected during this task; follow the actual editor's instructions before replacing files.

Article and author

Field	Entry
Article type	Research Article
Full title	Source-cohort expansion improves weaker-class coverage in cross-subject motor-imagery EEG decoding
Short title	Source-cohort expansion and class coverage in motor-imagery EEG
Given name / family name	Ziyuan / Zhu
Sole and corresponding author	Ziyuan Zhu
Affiliation	College of Artificial Intelligence Medicine, Chongqing Medical University, Chongqing, China
Address	Jinyun Campus, No. 61 Daxuecheng Middle Road, Shapingba District, Chongqing 401331, China
Correspondence	zzy2630816871@gmail.com
ORCID	https://orcid.org/0009-0005-1153-4926
Keywords	motor imagery; EEG; cross-subject decoding; source-cohort expansion; source-only model selection; weaker-class recall; reproducibility

Use the abstract in the current `manuscript_en.md`. The author has confirmed **Given name: Ziyuan; Family name: Zhu**. These manuscript and document metadata are corrected accordingly. The public ORCID name fields were observed reversed in the archived 9 October evidence; the external ORCID profile has not been edited by this task. The correct author-supplied identifier starts with **0009**, not the 0000 typographical error in the supplied review report.

Financial disclosure

The author received no specific funding for this work.

Competing interests

The author has declared that no competing interests exist.

Data and code availability

Use Data_Availability.md (Data_Availability.md). It identifies the current three datasets and exact source/results versions. Historical analyses from other datasets remain separate and are not substituted for the present 18-channel source-expansion evidence.

Ethics and participant information

This study was a secondary analysis of public EEG recordings and recruited no new participants. The source cohorts included 9 BNCI2014-001 participants and 51 Yang participants; external evaluation included 85 Dreyer participants, yielding 145 registered participant entries across the source and external cohorts. The author reports access only to provider participant codes and basic metadata, without names, contact details or an identity linkage key. According to the author's current confirmation, the research administration office of Chongqing Medical

University (科研处) confirmed on **17 October 2026** that the secondary analysis of **BNCI2014-001, Yang and Dreyer** data was exempt from ethics review under **Article 32(1)** of the 2023 Measures for Ethical Review of Life Science and Medical Research Involving Human Participants.

The date above is specific to Paper1 and supersedes the earlier author-only assessment for this updated candidate. The supplied review report's 16 October date concerned Paper2. The author confirmation has been recorded; a signed institutional document has not been independently inspected and no approval/exemption number has been supplied. Do not enter an invented number or describe this as formal IRB approval. The separate Ethics_Author_Statement.md (Ethics_Author_Statement.md) is an author statement, not a university letter. Original collection ethics and consent provenance remain attributable to the provider publications.

CRediT contributions

Ziyuan Zhu: Conceptualization; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Visualization; Writing – original draft; Writing – review & editing; Project administration.

These roles follow the author's declaration that he independently completed the work. They do not imply collection of the third-party original recordings. AI assistance is disclosed separately; AI systems are not authors. The new manuscript update still requires the author's final review.

AI assistance

ChatGPT and OpenAI Codex assisted literature and source checks, manuscript drafting and language revision, analysis and verification code, figure and document preparation, and publication of research materials to GitHub. Numerical statements in the revised manuscript were checked against frozen results and archived independent audit reports. No new model fitting or prediction inference was conducted for this manuscript revision. The author is responsible for checking references, scientific interpretation and final content before submission. The author

previously identified ChatGPT GPT-6; exact historical model versions are not independently established by the repository records and are not invented here.

See the companion boundary document for the disclosure and figure-caption requirement. Do not replace substantive code/figure assistance with a grammar-only statement.

Prior dissemination and submission route

The manuscript and research materials have been publicly accessible on GitHub. Earlier author confirmations reported no journal or preprint-platform publication. The existing submission was reported as PONE-D-26-50597 on 9 October 2026. This candidate develops the same work for a possible permitted update; it is not a second concurrent submission. **AUTHOR ACTION REQUIRED:** confirm the current portal state, any actual editorial instructions and exclusive-submission status. The supplied independent assessment is an author-supplied review, not an authenticated PLOS decision letter or an invitation to revise. Provide related manuscript and previous-submission information if requested.

Reviewer preferences and publication charge

The author previously reported no opposed reviewers and intends to personally pay the publication charge if accepted. No payment, fee-assistance application or publisher agreement is executed by this package. Recheck any current conflicts if suggesting reviewers or editors.

Upload map

Portal item	Current file
Main manuscript	`../main_manuscript.docx`
Author review PDF	`../review_manuscript.pdf`; also inspect the portal-generated PDF
Cover letter	`Cover_Letter.docx`
Main figures	Separate `../figures/Fig*.tif` files matching the manuscript citations
S1 Appendix	`../S1_Appendix.pdf`
S1 Data	`../S1_Data.zip`
S2 Data	`../S2_Neural_Predictions.csv.gz`
S3 Data	`../S3_CSP_Predictions.csv.gz`
Human-participants or other reporting form	Update the current official form if required; the historical completed form does not automatically cover this source-expansion manuscript

The original results ZIP is 21,240,637 bytes and is linked from the public repository. Do not upload it as a single supporting file if the portal applies the current 20 MB supporting-file limit; use the compact S1 Data and public full archive link. Internal release checklists and working notes are not scientific supporting-information uploads.

If an editor has invited revision

Only after inspecting the real decision letter, add a point-by-point response to its actual comments, a marked-up manuscript and the clean manuscript. This package does not manufacture those comments, an invitation or a manuscript number.